

3.4.4 Classification of programming languages and translators

Content	Additional information
Know that there are different levels of programming language: • low-level language • high-level language. Explain the main differences between low-level and high-level languages.	Students should understand that most computer programs are written in high-level languages and be able to explain why this is the case.
Know that machine code and assembly language are considered to be low-level languages and explain the differences between them.	Students should be able to • understand that processors execute machine code and that each type of processor has its own specific machine code instruction set • understand that assembly language is often used to develop software for embedded systems and for controlling specific hardware components • understand that assembly language has a 1:1 correspondence with machine code.
Understand that all programming code written in high-level or assembly languages must be translated. Understand that machine code is expressed in binary and is specific to a processor or family of processors.	
Understand the advantages and disadvantages of low-level language programming compared with high-level language programming.	

Content	Additional information
Understand that there are three common types of program translator: • interpreter • compiler • assembler. Explain the main differences between these three types of translator. Understand when it would be appropriate to use each type of translator.	Students will need to know that: • assemblers and compilers translate their input into machine code directly • each line of assembly language is assembled into a single machine code instruction • interpreters do not generate machine code directly (they call appropriate machine code subroutines within their own code to carry out statements).